

CAP-ROUGE, QC, Canada

WMO: 711860

Lat: 48.374N

Lon: 70.535W

Elev: 7

StdP: 101.24

Time Zone: -5.00 (NAE)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.7	(c) -24.4	(d) -31.0	(e) 0.2	(f) -26.0	(g) -28.7	(h) 0.3	(i) -24.0	(j) 19.6	(k) -7.3	(l) 18.1	(m) -9.9	(n) 5.2	(o) 300	(p) 0.741

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	27.6	18.9	25.7	18.5	24.0	17.8	20.8	25.0	19.8	23.5	18.9	22.3	4.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.3	22.5	18.5	13.4	21.5	17.6	12.6	20.6	60.2	25.2	56.8	23.5	53.7	22.4	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.8	(b) 15.4	(c) 13.5	(e) -30.0	(f) 31.3	(g) 2.6	(h) 1.3	(i) -31.9	(j) 32.2	(k) -33.5	(l) 33.0	(m) -35.0	(n) 33.8	(o) -36.9	(p) 34.7		
(a) 16.8	(b) 15.4	(c) 13.5	(e) -30.4	(f) 22.8	(g) 2.6	(h) 1.4	(i) -32.3	(j) 23.8	(k) -33.8	(l) 24.6	(m) -35.3	(n) 25.4	(o) -37.1	(p) 26.5		

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-12.9	-11.6	-4.7	3.0	10.7	16.3	19.0	18.4	13.8	7.2	0.5	-7.4
(7)		DBStd	12.06	6.99	6.48	6.30	4.06	3.85	3.54	2.66	2.84	3.58	3.63	4.26	6.15
(8)		HDD10.0	2953	709	604	456	212	37	1	0	0	7	101	287	538
(9)		HDD18.3	5176	967	838	714	461	239	80	24	34	141	345	536	797
(10)		CDD10.0	929	0	0	0	1	59	191	279	261	123	15	1	0
(11)		CDD18.3	108	0	0	0	0	2	20	44	37	6	0	0	0
(12)		CDH23.3	512	0	0	0	0	26	132	196	136	22	1	0	0
(13)		CDH26.7	96	0	0	0	0	6	28	37	22	3	0	0	0
(14)	Wind	WSAvg	5.8	6.8	6.4	6.4	6.4	5.6	5.0	4.4	4.2	4.9	6.1	6.5	6.6
(15)	Precipitation	PrecAvg	958	57	53	52	56	82	99	114	103	100	93	71	77
(16)		PrecMax	1165	114	108	105	94	163	158	266	240	219	155	136	143
(17)		PrecMin	810	8	23	19	20	34	43	38	41	54	27	23	38
(18)		PrecStd	92	24	23	23	21	30	31	47	42	33	32	22	22
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	3.9	3.6	8.7	16.7	26.9	29.8	30.0	29.2	25.9	19.4	13.0	4.9	
(20)		0.4% MCWB	2.5	2.0	4.0	8.8	16.2	19.3	20.8	20.5	19.2	14.6	10.4	3.2	
(21)		2% DB	1.5	1.5	6.0	12.8	22.1	26.7	27.4	26.7	22.8	15.8	8.9	2.9	
(22)		2% MCWB	0.9	0.4	2.6	6.7	13.3	17.7	19.3	19.7	17.7	12.1	7.1	1.8	
(23)		5% DB	-0.5	0.1	4.2	10.2	19.4	24.5	25.5	24.6	20.7	13.8	7.2	1.6	
(24)		5% MCWB	-1.2	-0.7	1.8	5.4	12.0	16.7	18.8	18.7	16.4	11.0	5.6	0.6	
(25)		10% DB	-3.0	-1.6	2.8	8.1	16.8	22.3	23.7	23.0	18.8	12.0	5.7	0.5	
(26)		10% MCWB	-3.8	-2.3	1.0	4.3	11.0	15.7	18.1	17.9	15.4	9.8	4.1	-0.3	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	3.3	2.5	5.2	9.4	17.4	21.1	22.3	22.2	20.6	15.2	11.1	3.5	
(28)		0.4% MCDB	3.9	3.2	7.1	13.7	24.7	27.2	25.8	26.0	23.8	18.4	12.6	4.6	
(29)		2% WB	0.9	0.6	3.2	7.2	14.7	19.3	20.9	20.8	18.5	12.9	7.6	2.0	
(30)		2% MCDB	1.5	1.2	5.0	11.1	19.4	24.1	24.9	24.7	21.7	15.0	8.6	2.7	
(31)		5% WB	-1.1	-0.7	2.1	5.9	13.1	18.1	19.9	19.7	17.1	11.4	5.8	0.8	
(32)		5% MCDB	-0.4	0.0	3.8	9.0	17.9	22.3	23.5	23.2	19.8	13.2	7.0	1.4	
(33)		10% WB	-3.6	-2.3	1.0	4.8	11.7	17.0	18.9	18.8	15.8	10.0	4.3	-0.2	
(34)		10% MCDB	-3.0	-1.6	2.6	7.4	16.2	21.1	22.3	21.8	18.1	11.6	5.5	0.4	
(35)	Mean Daily Temperature Range	MDBR	7.7	8.9	8.3	7.6	9.5	9.2	8.2	7.9	7.3	5.5	4.6	5.6	
(36)		5% DB MCDBR	8.5	7.7	8.5	11.6	14.3	13.0	11.2	10.6	9.9	8.6	6.5	5.4	
(37)		5% WB MCWBR	8.1	7.2	6.4	6.9	6.9	5.8	5.0	5.1	5.7	5.9	5.5	5.1	
(38)		5% WB MCDBR	8.7	7.6	7.6	9.8	12.2	10.5	9.4	8.8	9.1	7.9	6.3	5.3	
(39)	Clear-Sky Solar Irradiance	taub	0.268	0.290	0.323	0.395	0.374	0.395	0.418	0.397	0.381	0.338	0.313	0.274	
(40)		taud	2.271	2.217	2.184	2.134	2.372	2.349	2.301	2.362	2.389	2.490	2.411	2.310	
(41)		Ebn at Noon	800	861	886	850	887	867	840	843	820	807	744	743	
(42)		Edh at Noon	73	98	120	141	116	120	124	112	99	76	66	63	
(43)	All-Sky Solar Radiation	RadAvg	1.31	2.27	3.52	4.46	5.15	5.50	5.19	4.42	3.38	1.94	1.06	0.86	
(44)		RadStd	0.16	0.30	0.29	0.35	0.49	0.37	0.51	0.29	0.25	0.20	0.11	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page